

Applied Optimization

Exercise 8

Trust Region and Equality Constrained Optimization

September 18, 2025

What to hand in

A .zip compressed file renamed to `Exercisen-Group.zip` where n is the number of the current exercise sheet. It should contain:

- Hand in **only** the files you changed (headers and source). It is up to you to make sure that all files that you have changed are in the zip.
- A `readme.txt` file containing a description on how you solved each exercise (use the same numbers and titles) and the encountered problems. Indicate what fraction of the total workload each project member contributed.
Failure to do so will result in you losing a point.
- Other files that are required by your `readme.txt` file. For example, if you mention some screenshot images in `readme.txt`, these images need to be submitted too.
- For the theory exercise, put `TheoryExercise.pdf` with your solutions in the same .zip you submit for the code.
- Submit your solutions to ILIAS before the submission deadline. Late submissions will receive 0 points!
- **Important:** Your submission MUST compile without any issue in order to obtain a passing grade.

Trust Region Subproblem: Exact Solution (1 + Bonus 1 pts)

Consider the following trust region subproblem in two dimensions, $v \in \mathbb{R}^2$:

$$\begin{aligned} &\text{minimize } m(v) = g^T v + \frac{1}{2} v^T B v \\ &\text{subject to: } \|v\| \leq \Delta \end{aligned}$$

as discussed in [Nocedal, Sec 4.3] and the slides. In every problem, given the inputs B , g and Δ , solve the task and plot the quadratic model and the trust region with the **tool**.

Problem 1:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1/2$$

Compute an approximation of v^* with Cauchy point method.

Problem 2:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1/2$$

Compute an approximation of v^* with the Dogleg method.

Problem 3:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1$$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

Problem 4:

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Delta = 1/2$$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

Problem 5 (bonus):

$$B = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad \Delta = 1/2$$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

Problem 6: (bonus)

Consider now the the indefinite matrix

$$\tilde{B} = \begin{pmatrix} 4 & 0 \\ 0 & -2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \Delta = 1$$

Compute the λ^* and v^* which satisfy the KKT optimality conditions.

Problem 7: (bonus)

Same parameters as in problem 6, but now the gradient is zero $g = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Programming (1 pt)

Equality Constrained Newton Method

In previous exercises, we solve the mass spring system with constrained spring elements by adding some penalty terms to the objective function

$$E = \sum_{e(a,b)} E_{a,b} + \sum_{n \in C} \frac{1}{2} \text{penalty}_n ||(\mathbf{x}_n - \mathbf{p}_n)||^2$$

In this exercise, we instead solve an equality constrained optimization problem to achieve the same result. The problem can be described as following:

$$\begin{aligned} \text{minimize} \quad & E = \sum_{e(a,b)} E_{a,b} \\ \text{subject to} \quad & Ax = b \end{aligned}$$

It can be solved by the equality constrained Newton method. The newton step Δx can be computed by solving the KKT system as follows:

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \nu \end{bmatrix} = \begin{bmatrix} -\nabla f(x) \\ 0 \end{bmatrix}$$

You could refer to the lecture slides for the pseudo code “Newton’s Method with Equality Constraints”.

The four corner nodes in the spring graph are constrained to certain positions by specifying the equality constraints $Ax = b$. The code is provided in `setup_linear_equality_constraints(...)` function in the `MassSpringSystemT_impl.hh` file. Implement the algorithm in the `solve_equality_constrained(...)` function in the `NewtonMethods.hh` file.

Note that the equality constrained Newton works only when it starts from a feasible point which means the equality constraint is satisfied. Otherwise a projection operation that projects an infeasible point to the feasible set is needed. Implement it in the `project_on_affine(...)` function.

Hint: You can project your infeasible starting point x_0 to the hyperplane $Ax = b$ by finding a δx such that $A(x_0 + \delta x) = b$. Ask yourself what system you could solve to find such a δx

You can test your implementation by running the `EqualityConstrainedNewton` executable on random mass spring systems and visualize the results on the [usual website](#). As usual, you will find helpful unit tests alongside the main executable.